

Redaction gone wrong

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Now you DON'T see me.

This report has some critical data in it, some of which have been redacted correctly, while some were not. Can you find an important key that was not redacted properly?

Hints

- How can you be sure of the redaction?
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Tools Used

- `file` – Identify the actual file type using magic bytes
 - `strings` – Extract readable text strings from binary data
 - `grep` – Search for patterns (like “picoCTF” flag format)
 - `exiftool` – Examine metadata embedded in files
 - `pdftinfo` – Display PDF metadata and structure information
 - **Microsoft Word** – Open and inspect the document (since the PDF was created from a Word document)
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Complete Solution

Step 1: Download and Prepare

Download the file named `Financial_Report_for_ABC_Labs.pdf` from the challenge page.

Financial Report for ABC Labs, Kigali, Rwanda for the year 2021.

[REDACTED] - Just painted over in MS word.

[REDACTED]

Cost Benefit Analysis

Credit Debit

[REDACTED]

Expenses from the [REDACTED]

[REDACTED]

Redacted document.

At first glance, this appears to be a standard PDF financial report. However, the hint suggests that some redactions were done incorrectly.

Step 2: File Type Detection

First, verify the file type:

```
file Financial_Report_for_ABC_Labs.pdf
```

Expected output:

```
Financial_Report_for_ABC_Labs.pdf: PDF document, version 1.7, 1 page(s)
```

The file is a genuine PDF document.

Step 3: Search for Embedded Text

The flag might be stored directly within the file as readable text. Extract all printable strings and filter for the picoCTF flag format:

```
strings Financial_Report_for_ABC_Labs.pdf | grep -i "picoCTF"
```

Output: (no results)

⚠ The flag is not directly visible as plain text in the binary data.

Step 4: Examine PDF Metadata

Use `pdftinfo` to gather more information about the document:

```
pdftinfo Financial_Report_for_ABC_Labs.pdf
```

Output:

```
Title:           Microsoft Word - Financial Report for ABC Labs.docx
Creator:         Word
Producer:        macOS Version 11.4 (Build 20F71) Quartz PDFContext
CreationDate:    Sat Jul 17 21:44:11 2021 SAST
ModDate:         Sat Jul 17 21:44:49 2021 SAST
Custom Metadata: no
Metadata Stream: no
Tagged:          no
UserProperties:  no
Suspects:        no
Form:            none
JavaScript:      no
Pages:           1
Encrypted:       no
Page size:       612 x 792 pts (letter)
Page rot:        0
File size:       34915 bytes
Optimized:       no
PDF version:     1.7
```

Key finding: The document metadata reveals:

- **Title:** Microsoft Word - Financial Report for ABC Labs.docx
- **Creator:** Word

This indicates that the PDF was **created from a Microsoft Word document** (`.docx` file). The redactions were likely applied in Word before exporting to PDF.

Step 5: Understand the Redaction Problem

The PDF report has **black highlighted text** – this is how the information was “redacted” (blacked out). However, in Microsoft Word, black highlighting does not actually remove or encrypt the text. It simply changes the background color.

Why this is a problem:

- In a properly redacted PDF, the text should be completely removed or covered with a non-removable black box
 - In this case, the text is only **visually hidden** with black highlighting
 - If we open the original Word document (or the PDF in Word), the text can be revealed by changing the highlight color or copying the text
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Step 6: Extract the Flag by Copying Text

Since the PDF was created from Word, the underlying text is still present. We can simply **copy all the text** from the PDF and paste it into a text editor (Notepad, TextEdit, or any plain text editor).

Open the PDF in any PDF reader:

1. Select **Edit** → **Select All** (or `Ctrl+A` / `Cmd+A`)
2. Copy the text (`Ctrl+C` / `Cmd+C`)
3. Paste into a text editor (`Ctrl+V` / `Cmd+V`)

Result:

```
Financial Report for ABC Labs, Kigali, Rwanda for the year 2021.  
Breakdown - Just painted over in MS word.
```

```
Cost Benefit Analysis  
Credit Debit  
This is not the flag, keep looking  
Expenses from the  
picoCTF{<REDACTED>}  
Redacted document.
```

Key finding: The flag appears in plain text after copying!

Step 7: Alternative Method – Open in Microsoft Word

If you have Microsoft Word, you can also open the PDF directly (or convert it to Word format):

- 1. Right-click the PDF file → **Open with** → **Microsoft Word**
- 2. Word will convert the PDF to an editable document
- 3. Select all the text and change the **highlight color** or remove highlighting
- 4. The hidden text (including the flag) will become visible

This works because Word preserves the underlying text even when the PDF had black highlighting.

Final Result

picoCTF{<REDACTED>}

Note: The actual flag is redacted here. In the real challenge, copying the text from the PDF will reveal the complete flag.

Key Concepts

Concept	Description
False Redaction	Highlighting text in black does NOT actually redact it. The text remains in the document.
PDF from Word	PDFs created from Word retain the original text layer, even if it appears visually hidden.
Metadata Analysis	<code>pdftinfo</code> and <code>exiftool</code> reveal the source application (Word) and document properties.
Copy-Paste Attack	A simple copy-paste operation can bypass visual redaction because the text is still there.
Document Conversion	Opening a PDF in Word can reveal hidden text by allowing you to modify formatting.

Concept	Description
Proper Redaction	True redaction requires removing the text completely, not just covering it with a black box.

Pro Tips

- 1. **Never trust visual redaction** – Black highlighting or black boxes in Word/PDF documents can often be removed to reveal the underlying text.
- 2. **Check metadata first** – `pdfinfo` or `exiftool` can reveal if a PDF was created from Word, which often indicates poor redaction.
- 3. **Try copy-paste** – Before running complex tools, simply try to select and copy text from a PDF. You might be surprised what’s still there!
- 4. **Open in Word** – If you have Microsoft Word, opening a PDF there often strips away visual formatting and reveals hidden content.
- 5. **Proper redaction tools** – For real sensitive documents, use dedicated redaction tools that permanently remove text, not just cover it.